

Assignment 7:

Step 1: Problem Statement

This project's goal is to expand the functionality of the Virtual FedEx terminal created in Assignment 4 through the use of queues and priority queues. We will create 3 additional classes, Taxiways, Runway, and Air traffic controller, to simulate the assignment of planes from the tarmac, to the taxiway and then to the runway for take off. Certain planes will be given priority over others through the use of a priority queue. The results will be displayed throughout the progress to visually demonstrate the proper manipulation of the queues.

Step 2: What I know

In this case I'm already aware of all techniques applied in Assignment 4. We will continue to explore Has-A relationships within the new classes.

I'm also comfortable with the use of queues and priority queues.

Lastly, I should be good with all display methods.

Step 2: What I don't know

My only real challenge at this point will be the programming required to correctly use the `compareTo()` method in order to prioritize the planes by cargo appropriately.

Step 3: Pseudocode main & movePlanesTo taxiway

main

1 Create file objects

File filename = new File("FedExPlanes.txt");

File filename2 = new File("FedExTrucks.txt");

2 Read files

Scanner ~~readFileNextInt()~~; readfile = new Scanner

Scanner ~~readFileNextInt()~~; readfile2 = new Scanner

int planeCount = readfile.nextInt();

int truckCount = readfile2.nextInt();

Create terminal obj myterminal

Create objects (Plane)

while readfile.hasNext()

int start

int flight

double capacity

String cargo

String destination

Step 3: (cont)

```
create new CargoPlane myPlane (flight, cap, cargo, dest);  
myTerminal.addCargoPlane (stand, myPlane);  
end while
```

Create trucks

```
while readfile 2 ! empty
```

```
int dock
```

```
int truck
```

```
string dest
```

```
new SemiTruck myTruck (truck, dest);
```

```
myTerminal.addSemi (dock, myTruck
```

```
end while
```

move Planes to taxiway (Cargo Term Taxiways)

Call ~~the~~ Traffic Controller to move
planes to Taxiway using queue & Pri queue
and display result

* Display Empty Tarmac

Call move Plane to runway (Taxiways, Runway)

To move planes to runway based on Priority
and display results.

~~Call~~

Call cleared for takeoff (Runway) to
launch planes and display results

Class is Empty () for runway, Return T/R
and print result.

Step 3: (cont)

public void movePlanes to Taxiway (terminal, taxiway)

Loop Through stands $i=0 < \text{stands}$ $i++$

if stand ! null

nothing

else if

~~Cargo~~ = ~~military~~
~~Cargo~~ = ~~perishable~~
~~Cargo~~ = ~~medical~~

CargoCargo = military or perishable or medical

call compareTo() in CargoPlane

add to (offer) to Pri queue

else

add to (offer) to reg. queue

end if

end Loop

Step 4: What I Learned.

Learned that simply overriding the compareTo() method in the correct class will, if properly coded, prioritize the priority queue which is associated to it. That took me several hours of over thinking before it worked. Good Lesson.